CASE STUDY: Library Management System

Referring to the case of 'Library management system' introduced in the first chapter, let us develop this system on Waterfall approach.

In Waterfall model, first stage is preliminary analysis, which deals with the study of the current system, finding problems and establishing whether the new system will benefit the organization. In this library presently all transactions are done manually. Each member is allowed a certain number of books to borrow. He/she has cards to borrow books. Against each card a member can borrow one book from library.

Whenever a member wishes to issue a book and there are spare cards, then the book is issued.

Otherwise that request is not entertained. The numbers of members are increasing day by day. It is becoming a problem to manage the cards of members. So an automated system is required that can maintain the details about the books borrowed by different system. As we have studied the system, we now know about the existing problems. And we can positively establish the need for the automated record keeping system. After this preliminary investigation, second phase of Waterfall model, i.e., analysis phase begins.

In analysis, a detail study of system is performed. Each operation is looked into more details. From the preliminary analysis we know that the members are increasing and managing their cards is now difficult task for the library staff.

Let us find out why increasing members is a problem. There might be a case when a card gets misplaced either by library staff or by member itself. If this is indeed the case then a duplicate card is made. But a member can lie about it and can make a duplicate card even if the card is not lost. In that case that particular member is having more than maximum allowed cards. There is no means to identify such members.

Presently there is a need to have a system that can record the details about the books issued to members. Card system needs to be discarded. To solve this situation our library needs a computer application that has a database that contains details corresponding to each book issued. It should also have facilities to check if the number of books issued to a particular member doesn't exceed the maximum books allowed to him/her. Now we know what type of system is required, we can move to the next stage of waterfall model that is design stage. From analysis phase it is clear that we need a database to implement our new system.

In design stage it is decided as to which type of database will be used in system. Secondly we'll identify what data should this database store and in what format. After that various operations like issuing and returning books are finalized. Various checks like number of books issued not to exceed the maximum number are finalized. Various interfaces that are to be used for input are decided.

Once all these details are finalized, these are properly documented .These documents are used in building the system during the development stage, which follows the design stage. Using the design details the new system is built. Only things that are identified during the design stage are incorporated into the system.

There is no deviation from the design specifications. Suppose in the design phase the database to be used in the system is decided to be Oracle RDBMS then during development of the system only Oracle database is used. Similarly each design specification is taken care of and whole new system is build.

Now we have developed our new system. But before it is implemented at our user's site it needs to be tested for accuracy. So the system is tested to determine if it is performing correctly according to the requirements identified during the analysis phase. Each function of the system is tested. For example, we have issue book and return book functions. In return function it is checked if it is incrementing the count of variable that signifies how many books that member can issue more. In the issue function, it should be tested that this variable is decreased by one unit. Similarly whole of the system is tested.

After testing the system, it is implemented at the user's site. Now we have tested our new system for the library, it is implemented at the user's site i.e., library in our case.

After implementation, the systems require time to time maintenance. Maintenance can be for the software and hardware. Suppose two years after the implementation of the system, there occurs an alarming increase in number of members of the library. A situation can arise when all the memory for keeping the details have been exhausted then there will be a need to increase the memory of the system. Similarly the speed of processing requests might slow down. Then there will be a need for a faster processor. All these issues are maintenance issues.

So maintenance, though it is final stage of the system development, is equally important. In this section we explored how we can solve our case study problem 'Library management system' using traditional Waterfall model. Using Waterfall model is not mandatory. It is possible to follow other approaches to solve the same problem. It depends upon the developer, which approach to follow. In the next section, we illustrate case study using another approach i.e. Prototyping. This is an iterative approach towards system development.